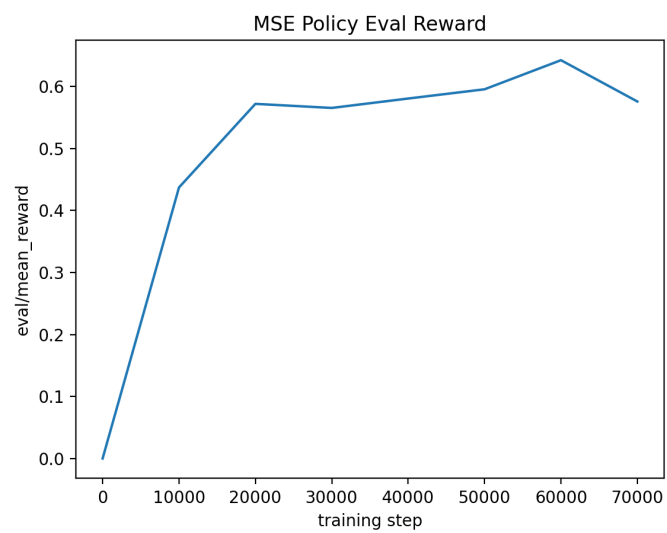
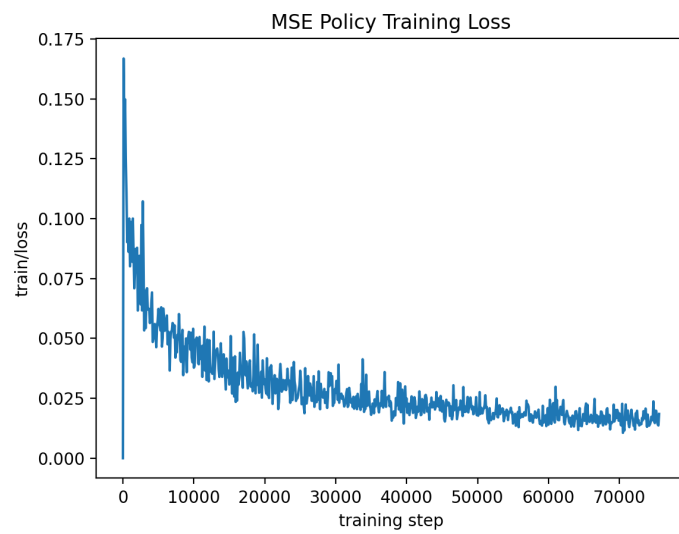


1. Action Chunk with MSE Loss

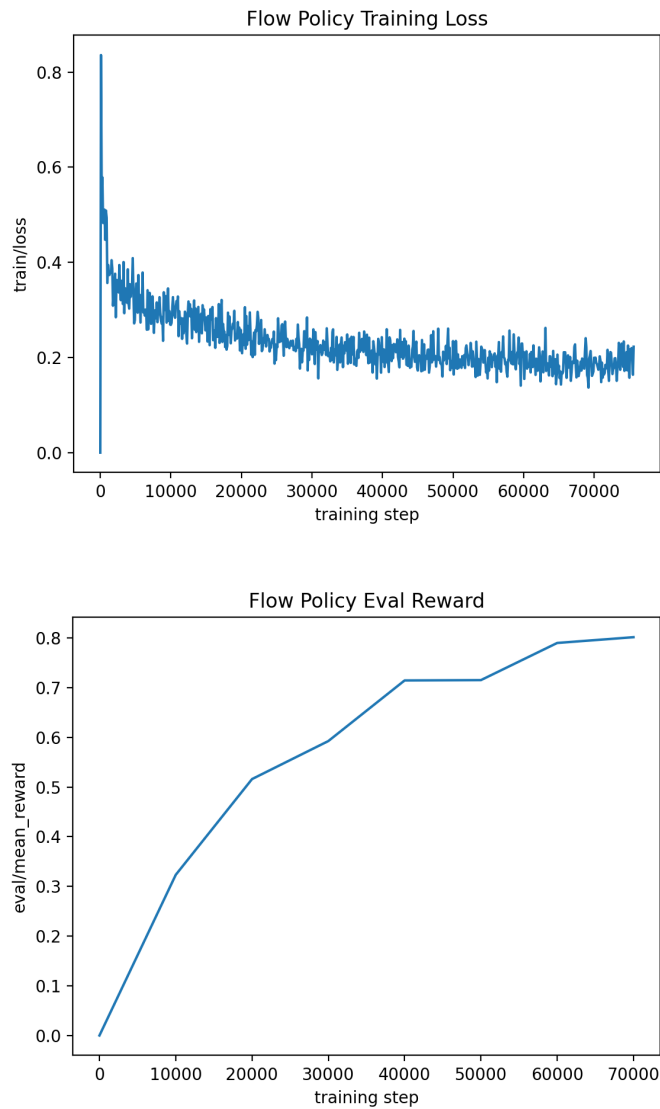
a) Training Curves:



b) MLP Architecture: My implementation adopts a 3 layer MLP, with 256 hidden dimensions respectively, and the activation function is ReLU.

2. Action Chunking with Flow Matching

a) Training Curves:



- b) Qualitative comparison: I used the same MLP structure and training hyperparameters for both networks, but generally the FM policy outperforms the MSE policy in terms of evaluation reward and policy behaviors demonstrated in videos. Specifically, the FM policy tends to act more decisively in the video, pushing at the right position and right angle, while the MSE policy tends to generate reasonable action direction overall but deviates in detailed motion, causing the “T” to move back and forth around target position. This is because the FM policy behaves better in multimodal behaviors, which tends to output an action within demonstration distribution, while the MSE policy tend to output an action that corresponds to the average of possible optimal action distributions. What’s more, the MSE policy outputs a chunk of action all at once, while the FM policy drives the noise to optimal action distribution step-by-step, which ensures an iterative refinement of the action chunk, though consuming more computational resources.